# Software Project Plan: Scope, Timeline, and Team Roles

Application: AI Lecture Assistant System | Student Name: Krishnaprasad.S | Roll Number: 241201026

## 1. Executive Summary & Project Objectives

This document presents the detailed Software Project Plan for developing the **AI Lecture Assistant Application**. The project aims to deliver a cross-platform mobile and web solution that automates classroom lecture recording, transcribes audio to structured study notes, and generates key exam questions using advanced Natural Language Processing (NLP).

## 2. Project Scope Specification

### In-Scope Features

- **Audio Recording & Ingestion:** High-fidelity live classroom recording and multi-format audio file uploads (MP3, WAV, M4A).
- **Speech-to-Text & Summarization Engine:** Real-time/batch speech transcription with automated formatting into bulleted summaries.
- **Key Question & Emphasis Mining:** NLP algorithms detecting vocal emphasis, repeated terms, and instructor cues to auto-generate review questions.
- **User Dashboard & Search:** Searchable note repository indexed by subject, course code, and recording date.
- **Export & Sharing:** PDF, DOCX, and TXT export with cloud backup integration (Google Drive / OneDrive).

### Out-of-Scope Features

- **Live Video Translation:** Real-time multilingual video lip-syncing (deferred to Phase 2).
- **LMS Deep Integration:** Automated grading and full Canvas/Blackboard direct grade sync.

## 3. Team Roles and Responsibilities Matrix

| Role | Responsibilities | Assigned Resource | Key Deliverables |
|---|---|---|---|
| **Project Manager / Scrum Master** | Overall planning, sprint scheduling, risk management, stakeholder communication. | Project Lead | Project Charter, WBS, Sprint Backlog |
| **Lead UI/UX Designer** | Wireframing, user journey maps, mobile/web responsive design, usability testing. | UI/UX Specialist | Figma Prototypes, Design System |
| **AI / NLP Engineer** | Speech recognition pipeline, summarization model tuning, key question mining. | ML/NLP Specialist | NLP Processing Engine & APIs |
| **Backend Developer** | API architecture, database management, cloud hosting, authentication & security. | Backend Engineer | REST APIs, Database, Docker Setup |
| **Frontend Mobile Developer** | Cross-platform mobile client implementation, audio capture module. | Mobile Engineer | React Native Mobile Application |
| **QA & DevOps Engineer** | Automated model testing, CI/CD pipeline, performance/load testing, cloud deployment. | QA/DevOps Lead | Test Plan, CI/CD Pipeline, Release |

## 4. Work Breakdown Structure (WBS) & Sprint Schedule

| Phase / Sprint | Duration | Key Activities | Major Milestones |
|---|---|---|---|
| **Sprint 1: Initiation & Requirements** | Weeks 1–2 | Requirement gathering, SRS preparation, tech stack selection. | Approved SRS & Architecture |
| **Sprint 2: UI/UX & Architecture** | Weeks 3–4 | Figma wireframing, DB schema design, API contract definition. | Interactive Prototypes & DB Schema |
| **Sprint 3: AI Engine Development** | Weeks 5–8 | Speech-to-text integration, NLP note summarization & question mining. | Functional AI Processing Pipeline |
| **Sprint 4: App & Backend Dev** | Weeks 9–12 | Mobile UI implementation, audio recorder, cloud integration. | Beta Mobile & Web Application |
| **Sprint 5: Testing & Refinement** | Weeks 13–14 | Integration testing, NLP accuracy tuning, user acceptance testing. | Bug Freeze & Quality Sign-off |
| **Sprint 6: Deployment & Handover** | Weeks 15–16 | App store release, AWS cloud deployment, user documentation. | Live App Production Launch |

## 5. Timeline & Milestone Gantt Chart Breakdown

| Project Milestone | Start Week | End Week | Duration | Status Track |
|---|---|---|---|---|
| **Requirements & SRS** | Week 1 | Week 2 | 2 Weeks | Completed |
| **UI/UX Design & Prototyping** | Week 3 | Week 4 | 2 Weeks | Completed |
| **Speech-to-Text & NLP Model Setup** | Week 5 | Week 8 | 4 Weeks | In Progress |
| **Mobile & Backend App Development** | Week 9 | Week 12 | 4 Weeks | Planned |
| **System Integration & QA Testing** | Week 13 | Week 14 | 2 Weeks | Planned |
| **Deployment & Launch** | Week 15 | Week 16 | 2 Weeks | Planned |

## 6. Resource Allocation & Budget Estimate

- **Human Resources:** 1 PM, 1 UI/UX Designer, 1 AI/NLP Engineer, 1 Backend Developer, 1 Mobile Developer, 1 QA/DevOps Engineer.
- **Infrastructure & Cloud Tools:** AWS S3/EC2 hosting, Speech API processing credits, Docker containerization tools.
- **Software Licensing:** Figma Team, GitHub Enterprise, Jira Software.

## 7. Risk Assessment and Mitigation Plan

| Risk Category | Identified Risk | Impact | Likelihood | Mitigation Strategy |
|---|---|---|---|---|
| **Technical** | Inaccurate speech recognition due to background noise/ accents | High | Medium | Implement noise-reduction pre-processing & model fine-tuning with acoustic datasets. |
| **Schedule** | Delays in NLP summarization model optimization | Medium | Medium | Use pre-trained Whisper/Transformer models and fine-tune incrementally. |
| **Operational** | Cloud server latency during peak lecture hours | High | Low | Deploy auto-scaling groups on AWS and implement asynchronous audio batch queue processing. |

## 8. Conclusion & Sign-Off

The structured 16-week Agile software project plan provides a robust blueprint for delivering the AI Lecture Assistant Application on time and within budget. Clear team roles, milestone-driven sprints, and proactive risk mitigation ensure seamless execution from initial requirements to final release.